FROM TREES TO TECH · ACTIVITY TTB-01 · FIELD SCIENCE

Root Grip Erosion Rescue

Can roots hold a slope together during a storm?

Win condition: Most soil retained after standard pour.

THE SCIENCE YOU ARE USING

Roots anchor soil. Bare soil on a slope washes away fast when rain hits it. Roots grip the soil and hold it in place, and a denser, deeper root network holds better. This is why planting vegetation is a real tool for stopping erosion.

Slope and runoff. Steeper slopes and bare surfaces shed water and soil quickly. You build a mini slope with root-like anchors, pour a standard amount of water, and measure how much soil stays put.

HOW TO BUILD AND RUN IT

1. **Build a bare slope.** Pack soil into a tray at a set angle with no roots. This is your control.
2. **Add root anchors.** Push string or yarn roots into a second slope to mimic a planted hillside.
3. **Pour the standard storm.** Pour the same measured water on each slope from the same height.
4. **Measure soil retained.** Compare how much soil washed out of each tray. Record the difference.
5. **Redesign the root network.** Add more or deeper roots and re-test to retain more soil.

Safety: Use trays and wipe spills. Allergy: Dust/pollen sensitivity.

MATERIALS

Trays	per team
Soil	shared
String or yarn roots	shared
Craft sticks	shared
Watering cups	per team

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why):

The ONE variable we are testing:

Baseline result:

After redesign:

DEFEND YOUR DESIGN

What evidence made you change your design?

If you ran it again, what is the ONE thing you would change?

HOW YOU SCORE (100 POINTS)

Scoring criterion	Points
Soil retained	40
Runoff control	25
Design explanation	20
Cleanup	15
Total	**100**